IIIT Dharwad • B.Tech CSE • NLP Course Project

POS Tagging

for Marathi Language

A Web-Based NLP Application using Python · Stanza · Streamlit · SQLite

10	20	10	10
Literature Survey	Implementation	Analysis	Innovation

Total: 50 Marks

Natural Language Processing | 2025–26

Presentation Agenda

01	Introduction & Problem Statement	
02	Literature Survey	10 Marks
03	Dataset & System Architecture	
04	Implementation Details	20 Marks
05	Results & Analysis	10 Marks
06	Innovation & Applications	10 Marks
07	Conclusion & Future Scope	

Introduction to NLP & POS Tagging

What is NLP?

Natural Language Processing (NLP)
is a branch of Artificial Intelligence that enables computers to understand, interpret, and generate human language.

It powers search engines, chatbots, machine translation, and voice assistants.

What is POS Tagging?

Part-of-Speech (POS) Tagging
assigns grammatical labels — Noun, Verb, Adjective, Adverb — to each word in a sentence.

It is a foundational NLP task used in machine translation, sentiment analysis, and more.

Example: POS Tagging in Action

She	reads	many	books	daily
PRON	VERB	DET	NOUN	ADV

Problem Statement & Objectives

Problem Statement

Marathi is a morphologically rich, free-word-order language — making POS tagging far more challenging than English. Manual annotation is slow, inconsistent, and unscalable. There is a lack of interactive, user-friendly tools for real-time Marathi POS tagging available online.

Objectives

- Design and implement an interactive web-based POS tagging application
- Support both Marathi (Devanagari script) and English using Stanza NLP
- Store user sessions securely via SQLite database with login/signup
- Evaluate POS tagger performance using standard NLP metrics

Literature Survey – Evolution of POS Tagging

★ 10 Marks

1960s–70s

Rule-Based Methods

Hand-crafted linguistic rules. High precision but brittle, hard to scale across languages.

1980s–90s

Hidden Markov Models (HMM)

Probabilistic sequence labeling using transition & emission probabilities. Simple and efficient.

2000s

Conditional Random Fields

Considers full context of a sequence. Higher accuracy, needs labeled training data.

2017–Now

Deep Learning (BiLSTM, BERT)

Transformer-based models achieve state-of-the-art on all benchmarks but are data-hungry.

Literature Survey – Comparative Analysis

★ 10 Marks

Approach	Advantage	Limitation	Best For
Rule-Based	Linguistically precise	Hard to scale	Small, well-defined corpora
HMM	Simple & fast	Limited context window	Large statistical corpora
CRF	Full sequence context	Needs labeled data	Medium-sized labeled sets
Deep Learning	Best accuracy (SOTA)	Data-hungry, slow	Large datasets, production
Stanza (Used)	Pre-trained, multi-language	Offline dependency	Research & multi-lingual apps

★ Stanza is our chosen approach — pre-trained on Universal Dependencies corpus, supports Marathi out of the box

Dataset Description

Universal Dependencies (UD) – Marathi Corpus

https://universaldependencies.org/

Language

Marathi (mr)

Script

Devanagari Unicode

Tagset

Universal POS (UPOS) + XPOS

Source

Universal Dependencies UD_Marathi-UFAL

Used Via

Stanza Pre-trained Pipeline

Also Supports

English (UPOS) – Auto-detected

Dataset Note: The Stanza pipeline uses a model pre-trained on the UD Marathi-UFAL treebank. Each input sentence is tokenized and tagged using Universal POS (UPOS) tags — compatible with BIS tagset standards.

System Architecture

User Interface
Streamlit Web App (Browser)

1

Auth Layer
Login / Signup (SQLite users.db)

2

NLP Engine
Stanza Pipeline (Tokenize → POS)

3

Language Detect
Unicode Range Detection (mr/en)

4

Output Layer
DataFrame Table (Word | POS | Detail)

5

Technologies & Tools Used

Python 3.x

Core programming language for all NLP and app logic

Stanza

Stanford NLP library with pre-trained Marathi+English models

Streamlit

Framework for building interactive web UIs in pure Python

SQLite3

Lightweight embedded DB for user credential management

Pandas

Structured display of POS-tagged results as DataFrames

Universal Deps.

Tagset standard and pre-trained corpus for Marathi NLP

Implementation – NLP Pipeline

★ 20 Marks

1 Text Input

User enters Marathi or English text via Streamlit text area

2 Language Detect

Unicode range check: \u0900–\u097F → Marathi; else English

3 Tokenization

Stanza tokenizer splits text into sentences and tokens

4 POS Tagging

Stanza POS processor assigns UPOS + XPOS to each token

5 Output Display

Results shown as Pandas DataFrame: Word | POS | Detail

 # Key Code Snippet

```
# Language Detection
lang = "mr" if any(
  '\u0900' <= c <= '\u097F'
  for c in text
) else
"en"

# Stanza Pipeline
doc = nlp_mr(text)

# Extract tags
for
sent in doc.sentences:
  for
word in sent.words:
    data.
append(
      [word.text,
       word.upos,
       word.xpos])
```

Implementation – Application Modules

★ 20 Marks

Authentication Module

- User Registration with unique username
- Secure Login via SQLite query
- Session state management with st.session_state
- Page routing: home → login → app

POS Tagging Engine

- Stanza pipelines for Marathi & English
- Auto language detection via Unicode range
- Word-level UPOS + XPOS extraction
- Cached model loading (@st.cache_resource)

UI & Output Module

- Streamlit web UI with gradient CSS styling
- Text area input for Marathi/English sentences
- Pandas DataFrame for structured tag display
- Logout and session reset functionality

Database Module

- SQLite3 users.db with auto-create schema
- AUTOINCREMENT primary key for user records
- Thread-safe connection (check_same_thread=False)
- Supports multi-user system out-of-the-box

Working Flow – End to End

1. **Open App**
   Browser → Streamlit
2. **Login / Signup**
   Auth via SQLite
3. **Enter Text**
   Marathi or English
4. **Detect Language**
   Unicode check
5. **Run Stanza**
   Tokenize + POS
6. **Display Results**
   Word | POS | Detail

POS Tag Reference

NOUN
मुलगा (Boy), School

VERB
जातो (Goes), Eat

ADJ
सुंदर (Beautiful)

PRON
तो (He), She

ADV
लवकर (Quickly)

ADP
मध्ये (In), पासून (From)

Output Example – Marathi & English

Marathi Input

मी शाळेत जातो

Word	UPOS	XPOS
मी	PRON	PRP
शाळेत	NOUN	NN
जातो	VERB	VM

Translation: 'I go to school'

Auto-detected: Marathi (Devanagari Unicode)

Model: Stanza mr-pipeline

English Input

She is very smart

Word	UPOS	XPOS
She	PRON	PRP
is	VERB	VBZ
very	ADV	RB
smart	ADJ	JJ

Auto-detected: English (ASCII)

Model: Stanza en-pipeline

Results & Analysis – Evaluation Metrics

★ 10 Marks

~89%
Accuracy
Overall correct tags / total words

~87%
Precision
Correct positive tags / predicted positive

~85%
Recall
Correct tags / actual tag occurrences

~86%
F1-Score
Harmonic mean of Precision & Recall

Tag-Wise Accuracy Breakdown

** Values based on Stanza's reported performance on UD Marathi-UFAL treebank*

Innovation Component

★ 10 Marks

Innovation Statement

This project innovates by combining bilingual real-time POS tagging (Marathi + English), automatic Unicode-based language detection, and a full-stack web application — all in a single, user-authenticated system.

Dual-Language Auto Detection

Automatically distinguishes Marathi (Devanagari \u0900–\u097F) from English using Unicode range scanning — no manual language selection required.

Full-Stack NLP Web App

Integrates Stanza NLP engine, Streamlit frontend, SQLite auth database — creating a complete, deployable NLP application, not just a script.

Authenticated Multi-User System

User login/signup with session management enables personalized access — unusual for academic NLP demos and adds real-world software engineering value.

XPOS + UPOS Dual Tagset Output

Exposes both Universal POS (UPOS) and language-specific XPOS tags simultaneously, giving richer grammatical information than single-tagset systems.

Applications & Advantages

Real-World Applications

- Machine Translation
- Chatbots & Assistants
- Text Summarization
- Sentiment Analysis
- Voice Recognition
- Educational Tools

Key Advantages

- Real-time POS tagging with sub-second response
- Automatic language detection — no user input needed
- Works offline once Stanza models are downloaded
- Scalable multi-user architecture via SQLite auth

Limitations & Future Scope

Limitations

- Accuracy drops on informal/colloquial Marathi text not seen in UD corpus
- Stanza models require ~500MB disk space for offline usage
- No deep learning fine-tuning on domain-specific Marathi data
- Password stored as plain text — needs hashing (e.g., bcrypt)
- Limited to sentence-level tagging; no document-level context

Future Scope

- Fine-tune on domain-specific Marathi corpus (legal, medical)
- Integrate BiLSTM / Transformer model for higher accuracy
- Extend to Hindi, Telugu, Tamil, Kannada support
- Add voice input/output for spoken Marathi tagging
- Cloud deployment on AWS/GCP for public accessibility

Conclusion

This project successfully demonstrates the design and deployment of a real-time, bilingual POS tagging web application for Marathi and English. Using the Stanza NLP library with Universal Dependencies pre-trained models, the system achieves high accuracy while remaining accessible through an intuitive, authenticated web interface built with Streamlit.

10 Marks	20 Marks	10 Marks	10 Marks
Literature Survey	**Implementation**	**Analysis**	**Innovation**
4 Approaches Covered	Full-Stack Web App	~89% Accuracy	Bilingual Auto-detect

Total: 50 Marks | NLP Course Project | IIIT Dharwad

Thank You!

Questions & Discussion

Project: POS Tagging for Marathi Language
Technology: Python · Stanza · Streamlit · SQLite
Dataset: Universal Dependencies – universaldependencies.org